



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study No.: 7

- 1. Case Study Title:** To demonstrate events handling in javascript, a) access and manipulate from elements using javascript. b) validate user input (eg. email, empty, fields) c) use various form events (submit, change, focus).

Case Study Program Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Shreya's (1196) - Event Handling</title>
<style>
body {
font-family: Arial, sans-serif;
background-color: #e8f0fe;
padding: 40px;
}
.form-box {
width: 450px; margin:
auto;
background-color: white; padding: 25px;
border-radius: 10px;
box-shadow: 0 4px 12px gray;
}
h2 {
text-align: center;
}
```



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```
label {
    display: block;
    margin-top: 15px;
    font-weight:
    bold;
}
input, select {
    width: 100%;
    padding: 10px;
    margin-top:
    5px;
    box-sizing: border-box;
}
input:focus,    select:focus
    { border:  2px  solid
    blue; outline: none;
}
.error {
    color: red;
    font-size: 13px;
}
button {
    width: 100%;
    margin-top:
    20px; padding:
    10px;
    background-color: #2563eb;
    color: white;
    border: none;
```



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```
#message {  
    text-align:  
        center;  
    font-weight: bold;  
    margin-top: 15px;  
}  
  
</style>  
</head>  
<body>  
<div class="form-box">  
    <h2>Shreya's student Registration Form</h2>  
    <form id="studentForm">  
        <label>Name:</label>  
        <input type="text" id="name">  
        <div id="nameError" class="error"></div>  
        <label>Email:</label>  
        <input type="text" id="email">  
        <div id="emailError" class="error"></div>  
        <label>Course:</label>  
        <select id="course">  
            <option value="">Select Course</option>  
            <option value="CSE">Computer  
Science</option>  
            <option value="IT">Information  
Technology</option>  
            <option value="ECE">Electronics</option>  
        </select>  
        <div id="courseError" class="error"></div>  
        <button type="submit">Submit</button>  
    </form>
```



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```
</div>

<script>

    // Access form elements
    let form =
document.getElementById("studentForm");

    let name = document.getElementById("name");
    let email = document.getElementById("email");
    let course = document.getElementById("course");
    // Focus event
    name.addEventListener("focus", function() {
        name.style.backgroundColor = "#eef4ff";
    });
    email.addEventListener("focus", function() {
        email.style.backgroundColor = "#eef4ff";
    });
    // Change event
    course.addEventListener("change", function() {
document.getElementById("courseError").innerText =
        "Selected Course: " + course.value;
    });

    // Submit event
    form.addEventListener("submit", function(event)
{
        event.preventDefault()
        ; let valid = true;
        // Name validation
        if (name.value.trim() === "") {
            document.getElementById("nameError").innerText
```



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```
        "Name cannot be empty";
        valid = false;
    } else {
document.getElementById("nameError").innerText =
"";
    }
    // Email validation
    let emailPattern =
/^^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (email.value.trim() === "") {
document.getElementById("emailError").innerText =
        "Email cannot be empty";
        valid = false;
    }
    else if (!emailPattern.test(email.value)) {
document.getElementById("emailError").innerText =
        "Enter a valid email";
        valid = false;
    }
    else {
document.getElementById("emailError").innerText =
"";
    }
    // Course validation
    if (course.value === "") {
```



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```
document.getElementById("courseError").innerText =
    "Please select a course";
    valid = false;
}
// Final result
if (valid) {
document.getElementById("message").innerText =
    "Registration submitted
successfully!";

document.getElementById("message").style.color =
"green";
}
else {

document.getElementById("message").innerText =
    "Please correct the errors.";
document.getElementById("message").style.color =
"red";
}
});
</script>
</body>
</html>
```



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3. Output:

Shreya's student Registration Form

Name:

Email:

Course:

Select Course

Submit

Shreya's student Registration Form

Name:

Email:

Course:

Computer Science

Selected Course: CSE

Submit

Registration submitted successfully!

4. Result/Conclusion:

Thus, JavaScript events were successfully handled using **focus**, **change**, and **submit**, and form elements were accessed, manipulated, and validated using JavaScript.